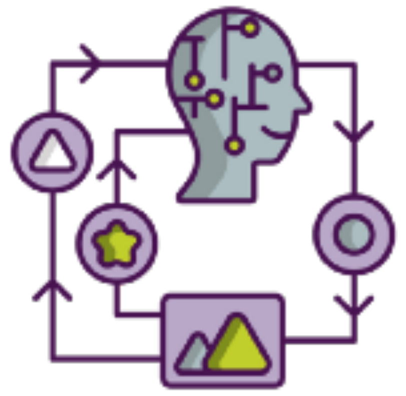
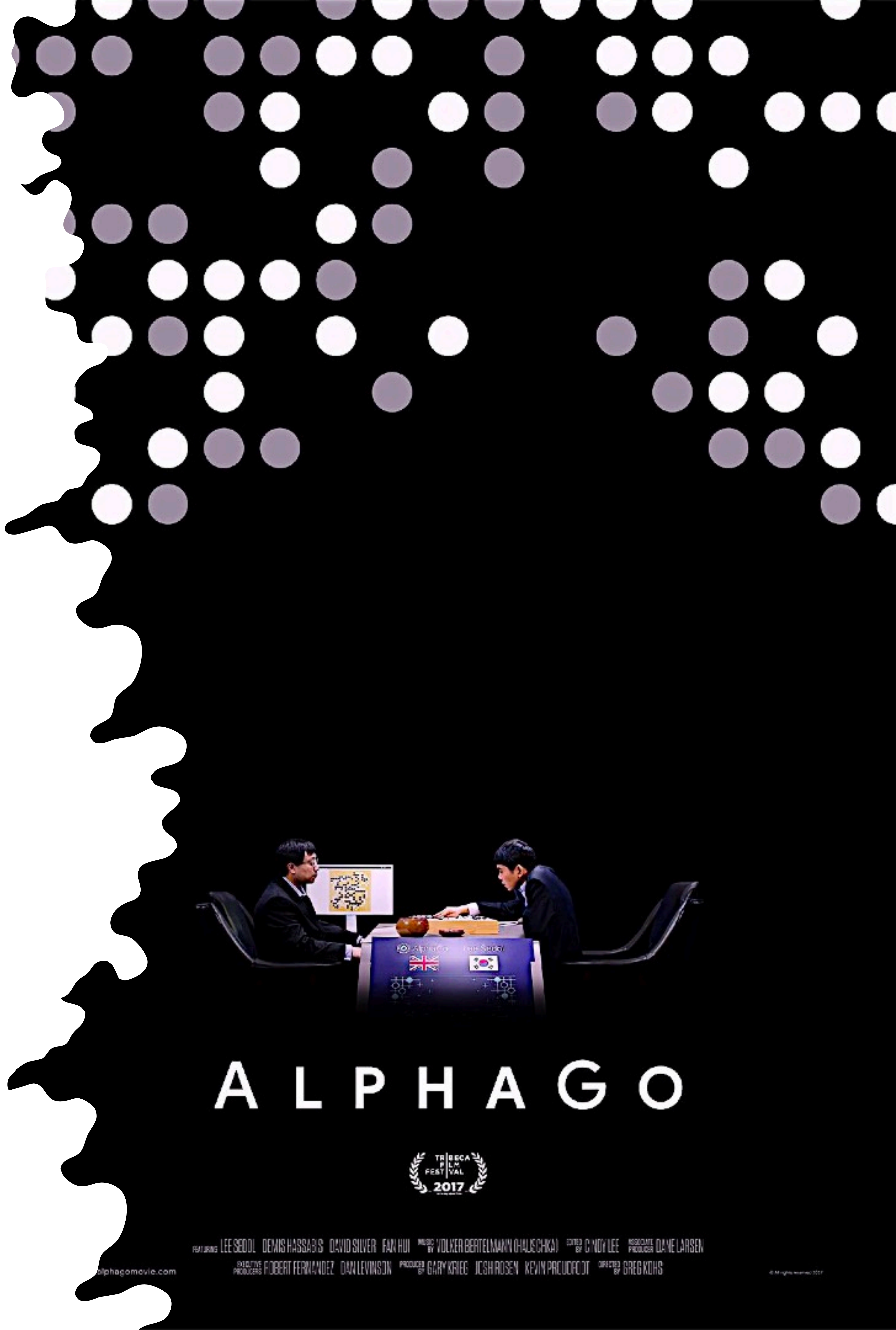
REINFORCEMENT LEARNING

Teaching through rewards for better decisions!



Reinforcement learning is a type of machine learning where an agent learns to <u>make decisions</u> by <u>interacting</u> with an <u>environment</u>, receiving rewards for good decisions and penalties for bad ones.



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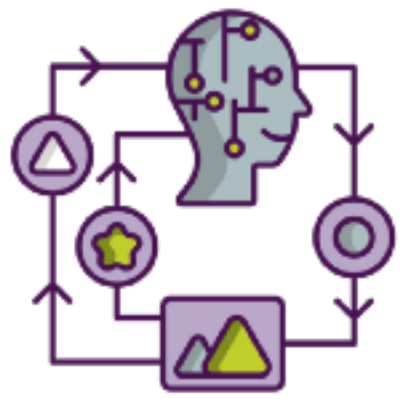

Reinforcement learning involves several key components: an agent, an environment, a set of states S, a set of actions A, a reward function $R(s, a)$, and a policy $\pi(a \mid s)$, which is a probability distribution over actions given states.

- Agent: The learner and decision maker.
- Environment: The external system the agent interacts with.
- State s: A situation the agent perceives.
- Action a: A move the agent makes.
- Reward R: A real number indicating the immediate benefit of performing action a in state s.
- Policy π: A mapping from states to actions, representing the agent's strategy.

The agent interacts with the environment in a series of discrete time steps. At each time step t, the agent receives a representation of the environment's state s_t, selects an action a_t, and sends this action to the environment. The environment responds by presenting a new state s_{t+1} to the agent and returning a reward r_t.

The agent's objective is to maximize the expected cumulative reward over time!

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